

PAPA, Hungary

WMO: 128250

Lat: 47.364N

Lon: 17.501E

Elev: 152

StdP: 99.51

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.5	-8.0	-14.2	1.1	-9.2	-11.3	1.5	-6.4	11.8	1.4	10.4	2.4	2.4	350	0.510

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	32.6	21.1	30.9	20.7	29.0	20.0	22.3	29.8	21.5	28.7	20.7	27.6	3.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.0	14.9	25.6	19.0	14.1	24.5	18.2	13.3	23.6	66.1	29.9	63.1	28.9	60.2	27.5	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	8.8	7.6	-13.9	34.7	3.5	2.2	-16.4	36.2	-18.4	37.5	-20.4	38.7	-22.9	40.3		
(5)			-14.6	24.0	3.6	0.8	-17.2	24.6	-19.4	25.1	-21.4	25.5	-24.0	26.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.0	0.0	1.9	6.0	11.5	15.5	19.4	21.2	20.7	16.1	11.1	6.4	1.4		
	DBStd	8.43	4.52	4.84	4.44	4.04	3.60	3.55	3.26	3.26	3.53	4.18	4.51	4.17		
	HDD10.0	1137	309	227	136	31	2	0	0	0	1	38	125	268		
	HDD18.3	2962	567	459	381	208	101	29	11	13	84	226	358	526		
	CDD10.0	1501	1	2	13	74	174	283	348	332	184	71	17	1		
	CDD18.3	285	0	0	0	2	14	62	101	87	17	2	0	0		
	CDH23.3	2976	0	0	0	25	169	647	1053	904	169	8	0	0		
	CDH26.7	1013	0	0	0	2	26	206	398	348	33	0	0	0		
(14) Wind	WSAvg	3.1	3.1	3.5	3.6	3.6	3.2	2.9	2.9	2.5	2.7	2.7	3.1	3.2		
(15) Precipitation	PrecAvg	673	35	30	38	47	62	77	70	72	61	49	54	53		
	PrecMax	994	70	69	86	101	199	161	149	211	157	108	97	97		
	PrecMin	447	4	3	7	5	18	8	10	2	18	1	8	5		
	PrecStd	134	17	20	23	25	38	37	34	49	43	30	21	26		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.2	15.9	20.2	26.1	29.0	33.1	34.9	35.4	29.8	24.5	19.5	14.8		
		MCWB	9.1	9.6	11.7	16.1	18.9	21.5	22.0	21.0	19.2	17.3	14.1	11.1		
	2%	DB	10.5	13.1	17.9	23.6	27.1	31.1	32.5	32.7	27.3	22.0	17.0	11.2		
		MCWB	7.4	8.5	10.8	14.4	17.9	21.1	21.1	21.0	18.6	16.0	12.7	8.4		
	5%	DB	8.2	11.1	15.7	21.3	25.2	29.0	30.8	30.5	25.2	20.0	15.0	8.8		
		MCWB	5.8	7.2	10.0	13.6	17.2	20.5	20.7	20.5	17.8	15.0	11.6	6.8		
	10%	DB	6.3	9.0	13.3	19.3	23.2	27.1	28.9	28.2	23.1	17.9	12.9	7.0		
		MCWB	4.5	6.1	8.6	12.4	16.1	19.3	20.0	19.6	16.9	14.0	10.2	5.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.5	10.1	12.7	16.7	20.2	23.2	23.5	23.2	20.5	17.9	14.4	11.0		
		MCDB	12.7	14.9	18.5	24.7	27.1	30.6	31.2	30.5	27.3	22.9	18.7	14.2		
	2%	WB	7.7	8.8	11.4	15.3	18.9	21.9	22.3	22.0	19.4	16.6	13.0	8.6		
		MCDB	10.3	12.6	16.7	22.2	25.4	28.9	29.8	29.8	25.5	21.0	16.5	11.1		
	5%	WB	6.0	7.6	10.2	14.1	17.7	20.9	21.4	21.2	18.5	15.5	11.8	6.9		
		MCDB	7.8	10.5	15.0	20.1	23.9	27.6	28.6	28.6	23.8	19.2	14.6	8.4		
	10%	WB	4.6	6.3	9.0	13.0	16.7	19.9	20.6	20.3	17.4	14.4	10.4	5.5		
		MCDB	6.2	8.6	13.0	18.4	22.2	26.1	27.5	26.9	22.0	17.4	12.6	6.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.0	7.6	9.6	11.0	10.9	11.4	11.9	12.0	10.6	9.5	6.8	5.6		
		MCDBR	8.4	11.2	13.3	14.5	14.1	14.0	14.8	15.5	13.9	11.9	9.3	8.1		
	5% WB	MCWBR	6.1	7.2	7.8	7.6	6.9	6.0	5.6	5.9	6.5	6.8	6.0	6.1		
		MCWBR	8.2	9.9	12.1	13.3	12.7	12.9	13.2	13.6	12.1	11.0	9.0	7.4		
(40) Clear-Sky Solar Irradiance	taub	taub	0.327	0.353	0.390	0.436	0.429	0.431	0.439	0.441	0.417	0.386	0.358	0.328		
		taud	2.363	2.319	2.237	2.127	2.201	2.234	2.213	2.219	2.253	2.317	2.346	2.378		
	Ebn at Noon	Ebn at Noon	742	799	821	815	837	836	823	803	783	748	694	694		
		Edh at Noon	73	93	118	144	139	135	136	130	115	93	74	64		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.09	1.91	3.11	4.54	5.42	5.89	5.78	5.12	3.66	2.33	1.23	0.84		
	RadStd	RadStd	0.11	0.27	0.38	0.44	0.53	0.41	0.44	0.50	0.42	0.23	0.12	0.10		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (74 neighbors)	+0.48	N/A	N/A	+0.91	+0.52	+0.46	-122	-173	+102	+57

Nomenclature: See separate page